

Table [climate-associated cluster candidates, full]. Candidate climate-associated LD clusters (primary analysis: Åland-excluded, LD-pruned Ω), with the eMLG gate. **Bold** = restricted set (eMLG $\text{BF}(\text{dB}) \geq 15$).

Cluster	Chr	Position (Mb)	Size	Axis	Sig.	% sig.	eMLG BF(dB)	eMLG β	XtX	DI%
F63613	Chr24	1.26–1.68	259	PC1	215	83.0	47.6	+0.098	38	61.4
F17706	Chr5	3.04–3.34	11	PC2	7	63.6	35.8	+0.123	25	27.3
F41740	Chr13	5.84–6.18	50	PC2	11	22.0	30.5	+0.114	26	0.0
F64080	Chr24	3.08–4.36	28	PC2	7	25.0	26.5	+0.122	26	3.6
F63780	Chr24	1.76–1.82	83	PC2	73	88.0	25.3	+0.117	26	0.0
F67245	Chr26	7.36–7.48	71	PC2	43	60.6	23.2	+0.104	22	62.0
F27572	Chr8	0.86–6.91	439	PC2	27	6.2	21.4	+0.068	24	0.0
F67234	Chr26	6.56–7.48	42	PC2	7	16.7	18.1	+0.084	21	11.9
F63847	Chr24	2.03–2.38	19	PC1	5	26.3	17.8	+0.058	23	15.8
F27537	Chr8	5.50–7.04	639	PC1	136	21.3	16.8	+0.062	24	38.8
F41775	Chr13	6.03–6.04	11	PC2	5	45.5	15.0	+0.091	22	9.1
F32704	Chr9	9.01–9.13	10	PC2	7	70.0	14.1	+0.097	25	20.0
F17649	Chr5	0.92–2.71	113	PC2	24	21.2	13.6	+0.094	19	13.3
F13273	Chr3	14.73–14.73	7	PC2	5	71.4	13.2	+0.087	21	0.0
F67274	Chr26	7.48–7.51	46	PC2	5	10.9	13.1	+0.088	23	34.8
F38791	Chr12	9.83–9.84	17	PC1	6	35.3	12.5	+0.061	22	5.9
F33634	Chr10	8.62–8.87	126	PC1	5	4.0	11.9	+0.053	23	24.6
F67010	Chr26	1.44–5.91	3587	PC2	135	3.8	11.2	+0.089	22	21.8
F48424	Chr15	8.19–8.21	19	PC1	5	26.3	10.8	+0.057	26	0.0
F12274	Chr3	9.04–12.97	210	PC2	6	2.9	10.8	+0.058	24	0.0
F14815	Chr4	6.45–11.04	99	PC2	22	22.2	9.8	+0.085	30	28.3
F67030	Chr26	1.55–4.72	393	PC2	33	8.4	9.6	+0.078	23	73.3
F17687	Chr5	0.89–4.27	734	PC1+2	20	2.7	7.6	+0.070	23	13.2
F51319	Chr17	0.05–4.95	3187	PC2	60	1.9	5.4	+0.059	19	0.0
F11512	Chr3	6.19–8.49	1143	PC2	43	3.8	3.7	+0.067	20	0.6
F13942	Chr4	5.45–5.60	33	PC2	5	15.2	2.4	+0.064	22	0.0
F13038	Chr3	13.37–13.54	103	PC2	12	11.7	0.4	+0.056	26	0.0
F17658	Chr5	0.90–3.97	139	PC1	5	3.6	-0.0	+0.037	20	7.9
F27536	Chr8	0.86–5.48	3587	PC2	5	0.1	-6.5	+0.034	21	0.9
F67057	Chr26	1.82–6.75	1282	PC2	18	1.4	-11.4	+0.017	22	39.9
F68300	Chr27	0.05–3.21	497	PC2	17	3.4	-11.7	+0.016	20	12.1
F13043	Chr3	13.22–13.94	93	PC1	5	5.4	-14.6	+0.009	25	0.0

Ordered by the cluster-level eMLG association on the axis the cluster entered on. Size = member markers; Axis = climate axis of entry (member-count); Sig. = member markers with $\text{BF}(\text{dB}) \geq 15$; % sig. = $\text{Sig.}/\text{Size}$; eMLG $\text{BF}(\text{dB})$ and β = the cluster's consensus-genotype association from a BayPass run on the eMLGs (primary config); XtX = mean among-population differentiation (a prerequisite for association, uniformly elevated); DI% = proportion of members with $\text{DI} > -25$. **Bold** clusters pass the eMLG gate ($\text{BF}(\text{dB}) \geq 15$; 11 of 32) and constitute the restricted candidate set; the rule separates them from clusters passing on member count alone. Genome-wide median cluster size = 9 markers.